WordPress, ImageMagick and libheif unci RCE

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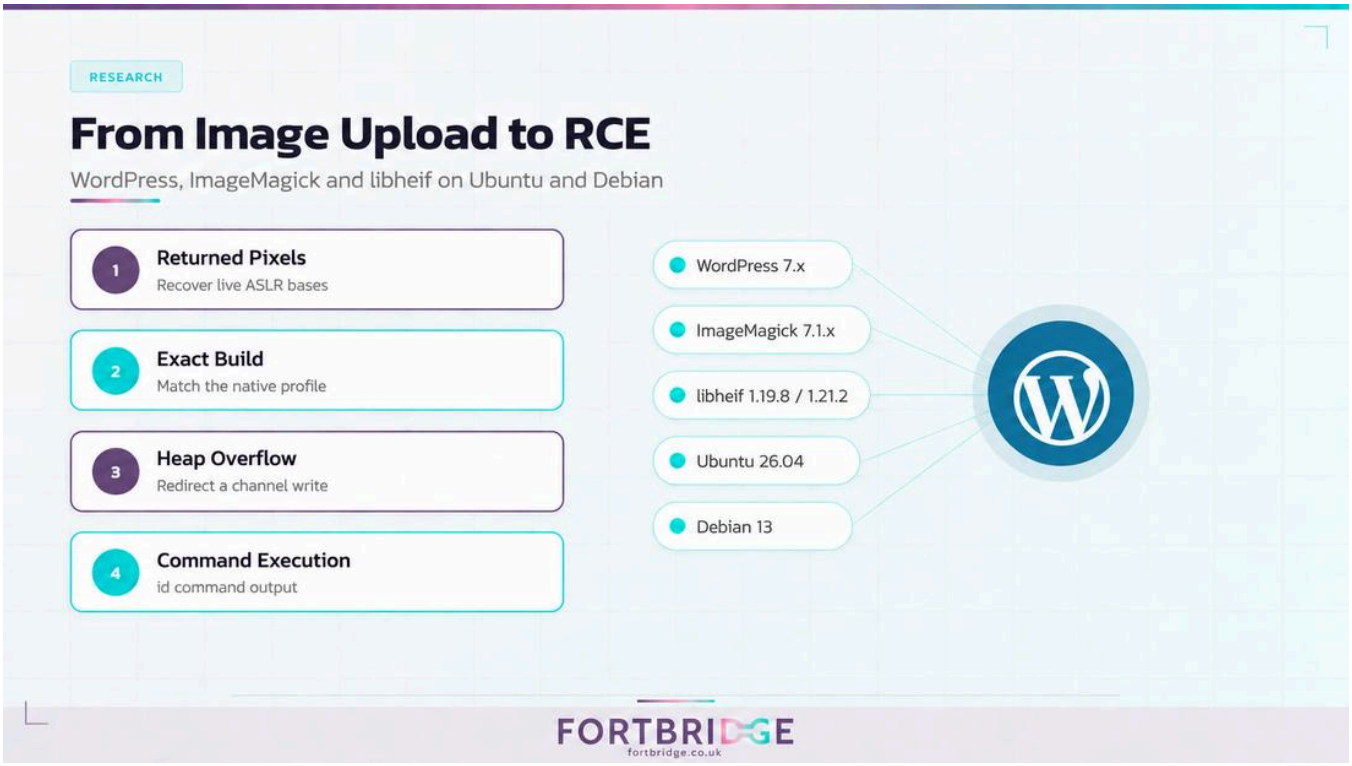
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`README.md`

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WordPress, ImageMagick and libheif `unci` RCE



This repository contains Fortbridge's proof-of-concept RCE chain for two exact WordPress, PHP-FPM, ImageMagick and libheif builds. It uses authenticated media uploads to identify the native stack, recover live addresses from pixels in WordPress image derivatives, build an ASLR-adjusted HEIC trigger, and verify the output of `id` with a separate HTTP request.

The chain combines two libheif vulnerability families:

- GHSA-x8r2-mggj-j6wr, a file-controlled heap overflow in mixed-interleave `unci` decoding.
- GHSA-2jg2-4ch7-h545, derived-image and plane-geometry flaws that can expose heap bytes in generated images.

Use this code only on systems you own or are explicitly authorised to test. Attempts can crash image-processing workers and run a command as the PHP-FPM account. Patch libheif instead of treating this proof of concept as a scanner.

Supported profiles

`run_rce.sh` supports these exact amd64 builds:

Profile	Native stack	Result
`ubuntu-26.04-amd64-wp-fpm-2026-09`	WordPress 7.1.1, PHP-FPM 8.5.4, ImageMagick 7.1.2.18, libheif 1.21.2, glibc 2.43	6 of 8 fresh parent processes within 8 positions
`debian-13-amd64-wp-fpm-2026-09`	WordPress 7.0, PHP-FPM 8.4.24, ImageMagick 7.1.1.43, libheif 1.19.8, glibc 2.41	22 of 24 fresh parent processes within 16 positions

These are exact package profiles, not operating-system guesses. A target with different native artifacts needs a separately measured and verified profile. The hardcoded build values, package hashes and GNU build IDs are in `profiles/build_profiles.json`.

How the chain works

1. The exploit signs in to WordPress and uploads disclosure images through the

normal media route.

1. It downloads WordPress-generated derivatives and decodes pointer

relationships from their pixels.

1. The selector accepts only one complete native-module pattern and one

unambiguous libc base. Missing, mixed, or conflicting evidence stops the run before a trigger is sent.

1. The selected profile supplies the exploit strategy, hardcoded offsets,

writable ranges, and retry limit for that exact native stack.

1. The exploit builds a request-specific `unci` image and sends it to the same

media route.

1. Success requires a separate HTTP GET to a path that started at 404 and now

returns `uid=33(www-data)`.

An upload returning HTTP 503 is evidence of a worker crash, not command execution. The Debian profile checks and then writes the proof to the fixed `/rce-proof.txt` path in the lab web root; a successful run leaves this file behind. Use only an isolated authorised lab and remove the proof file afterward.

Profile selection and ASLR

The HTTP `Server` header is used only to choose which harmless allocation schedule to try first. It never selects a profile.

Ubuntu identification uses an exact pair of pointers inside its libheif artifact. Debian identification can use a libheif pair and then recovers the MagickCore base from adjacent `CompareSplayTreeString` and `RelinquishMagickMemory` callback pointers. Each pointer must resolve to the same page-aligned base for the declared public binary.

These runtime addresses are recovered again for every fresh PHP-FPM parent process. Build-relative offsets remain constant only for the exact binaries listed in the profile manifest.

Ubuntu strategy

Seven long-running requests occupy seven workers in the eight-child FPM pool. The remaining worker performs calibration and handles the trigger. A two-tile disclosure returns two 64 MiB-aligned glibc arenas and decoder anchors. The exploit chooses the higher validated arena and builds a 1024 by 128 trigger.

The controlled Cr tail reaches a live channel vector. Later chroma writes are redirected into the predicted decoder, where they install a fake dispatch structure. A virtual call then reaches an alignment-safe `do_system` call site inside the exact Ubuntu libc build.

Debian strategy

The Debian trigger is 512 pixels wide. Source row 32 reaches the Cr allocation boundary. The row-32/33 bridge repairs vector fields for both observed pointer alignments, then redirects later plane writes into the writable MagickCore load after RELRO.

A benign 64-frame HEIC prepares ordinary MagickCore objects for the following leak attempt. The image does not contain the `unci` corruption primitive. The next returned derivative may expose the two callback pointers used to recover the live MagickCore base.

The implementation caches the selected libc and MagickCore bases only while the same PHP-FPM parent process is alive. If that process restarts, the exploit must recover both bases again.

Requirements

- Python 3.11 or newer
- Bash, `curl`, `jq`, and `sha256sum`
- Python packages from `requirements.txt`
- a WordPress account with the `upload_files` capability
- an explicitly authorised target matching a supported profile

```
python3 -m venv .venv
. .venv/bin/activate
python3 -m pip install -r requirements.txt
./prepare_payloads.sh
```

The large scheduling fixtures are generated locally and ignored by Git. `prepare_payloads.sh` verifies their SHA-256 hashes. Small checked-in fixtures are covered by `SHA256SUMS`.

Run the exploit

The repository contains no default target credentials.

```
export WP_URL='https://authorised-target.example'
export WP_USERNAME='author'
export WP_PASSWORD='replace-me'
./run_rce.sh
```

The optional first argument lowers the position limit but cannot exceed the bound declared by the selected profile:

```
./run_rce.sh 8
```

Successful output includes:

```
RCE_SUCCESS url=https://authorised-target.example/a
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

Reproducible lab containers

The lab images use immutable Ubuntu and Debian base-image digests, exact WordPress archives with SHA-256 checks, WP-CLI 2.12.0 with a SHA-256 check, and exact versions for the native packages that define each profile.

```
./scripts/build_ubuntu.sh
./scripts/build_debian.sh
./scripts/show_profiles.sh
./scripts/smoke_container.sh ubuntu 8877
./scripts/smoke_container.sh debian 8878
```

To keep a lab running:

```
docker run --detach --name libheif-unci-ubuntu \
  --publish 127.0.0.1:8877:80 \
  --env LAB_URL=http://127.0.0.1:8877 \
  libheif-unci-wordpress-rce:ubuntu
```

The local lab account is `author` with password `author-pass`. Those are test credentials for loopback containers, not target defaults in the exploit.

Repository layout

```
run_rce.sh                          profile-driven entry point
exploit/                            shared classifier and Ubuntu strategy
exploit/debian13/                   Debian strategy and payload builders
profiles/build_profiles.json        binary identities, offsets, and bounds
containers/                         exact Ubuntu and Debian lab definitions
scripts/                            container build and smoke-test helpers
tools/                              profile verification and regression checks
docs/                               technical notes and reliability results
evidence/                           reduced non-sensitive validation records
```

Verify a build profile

Extract the matching public packages, then map each artifact to its ELF file:

```
python3 tools/verify_build_profile.py \
  --profile ubuntu-26.04-amd64-wp-fpm-2026-09 \
  --artifact php_fpm=/extracted/usr/sbin/php-fpm8.5 \
  --artifact php_imagick=/extracted/usr/lib/php/20250925/imagick.so \
  --artifact imagemagick_heic_coder=/extracted/usr/lib/x86_64-linux-gnu/ImageMagick-
7.1.2/modules-Q16/coders/heic.so \
  --artifact magickcore=/extracted/usr/lib/x86_64-linux-gnu/libMagickCore-7.Q16.so.10.0.2
\
  --artifact libheif=/extracted/usr/lib/x86_64-linux-gnu/libheif.so.1.21.2 \
  --artifact libc=/extracted/usr/lib/x86_64-linux-gnu/libc.so.6 \
  --artifact libstdcxx=/extracted/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.35
```

The verifier checks package metadata, build IDs, file hashes, libc symbols, and instruction signatures. It does not qualify a new build automatically.

Regression checks

```
python3 -m unittest discover -s tests -p 'test_*.py'
python3 -m py_compile exploit/*.py exploit/generators/*.py \
  exploit/debian13/*.py exploit/debian13/generators/*.py tools/*.py tests/*.py
bash -n run_rce.sh prepare_payloads.sh exploit/debian13/*.sh containers/entrypoint.sh
scripts/*.sh
python3 tools/audit_supported_path_constants.py
sha256sum -c SHA256SUMS
```

See `docs/reliability.md` for the qualification method and `SECURITY.md` for reporting and safe-use guidance.

Evidence screenshots

Only the final publication evidence is kept in this repository:

- validated Ubuntu and Debian stacks
- controlled decoder virtual-table pointer
- ASLR values recovered from returned pixels
- successful command-execution proof

`docs/profile-constant-inventory.md`

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Supported profile constant inventory

The supported route has one selection boundary:

1. `run_rce.sh` obtains returned-pixel evidence.
2. `remote_profile_selector.py` requires one exact native stack and libc base.
3. `profile_driver.py` validates the selection and returns its declared

strategy, refresh policy and position bound.

1. The shell driver dispatches either the Ubuntu overlay strategy or the

Debian MagickCore strategy.

The HTTP `Server` value can order the two allocation schedules. It does not select a profile.

Shared profile facts

Fact	Manifest location	Supported consumer
Native binary identities	`native_stacks.*.artifacts`	`verify_build_profile.py`
Returned libheif patterns	`native_stacks.*.returned_pixel_fingerprints`	`remote_profile_selector.py`
Returned module-base patterns	`native_stacks.*.returned_pixel_module_bases`	selector and Debian sink policy
Native-stack to libc binding	`native_stacks.*.libc_profile`	selector and driver validation
Returned libc anchors	`libc_profiles.*.returned_pixel_leak`	`remote_profile_selector.py`
Driver strategy and refresh policy	`profiles.*.driver`	`profile_driver.py`, `run_rce.sh`
Qualification position bound	`profiles.*.qualification.max_positions`	`profile_driver.py`, `run_rce.sh`

Fact	Manifest location	Supported consumer
libc call and dispatcher offsets	`profiles.*.libc`	Ubuntu builder and Debian probe
Dispatcher object and slot ABI	`profiles.*.libc`	both trigger generators

Ubuntu-only profile facts

Fact	Manifest location	Consumer
Thread arena alignment and fingerprints	`profiles.*.thread_leak`	thread leak decoder and trigger validation
Thread-leak image geometry and byte positions	`profiles.*.thread_leak`	`decode_thread_arena_tiles`
Decoder and plane offsets	`profiles.*.early_worker`, `w1024_overlay_worker`	Ubuntu trigger builder
Trigger dimensions and scheduling	worker records	Ubuntu trigger builder
Fake-vtable and command placement	worker records	Ubuntu generator

Debian-only profile facts

The Debian strategy reads libc dispatcher, command-object and call-site offsets from `profiles.*.libc`. Its MagickCore callback offsets, RELRO boundary, writable load and permitted sink interval come from `native_stacks.*.returned_pixel_module_bases.magickcore`.

The top-level driver also passes the selected libc and MagickCore bases into the internal sweep. `exploit/debian13/probe.py` must not reintroduce the profiled libc constants as integer literals.

Code-owned structural constants

Profiles do not own universal file-format encodings, ELF parsing rules, canonical x86-64 pointer bounds, HTTP status codes, allocator chunk-field positions or ISO BMFF layouts. Those values describe the protocol or payload representation rather than one packaged native stack.

Regression guard

`tools/audit_supported_path_constants.py` checks the supported functions most likely to regress. It rejects measured profile values if they reappear as Python integer literals in the

thread decoder, Ubuntu trigger builder or Debian probe.

`docs/reliability.md`

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Reliability results

Updated: 2026-09-27

This document records the final qualification of the public `run_rce.sh` workflow. Older exploratory layouts and failed branches were removed because they are not operator guidance and are not used by the supported entry point.

What one result means

A PHP-FPM parent process creates and replaces the workers that handle image uploads. Uploads sent while one parent process remains alive are related, even when different workers handle them. A fresh run therefore means a new parent process with newly randomised addresses, followed by a fixed maximum number of candidate exploit positions.

A run counts as successful only when all of these conditions hold:

1. The selector identifies one exact native stack from returned image pixels.
2. The exploit recovers the required runtime bases.
3. The proof path returns HTTP 404 before the trigger.
4. The proof path later returns HTTP 200 with `uid=33(www-data)`.

HTTP 503, a connection reset, or a worker crash does not count as execution.

Final results

Outcome	Ubuntu 26.04	Debian 13
Fresh parent processes	8	24
Parent processes with execution	6	22
Maximum positions per parent process	8	16
Candidate positions	26	93
Verified executions	6	22
Triggers without a proof	19	33
Safe rejections before the trigger	1	38

Outcome	Ubuntu 26.04	Debian 13
Separate classification uploads	142	78
Safe classification rejections	134	54
Exact profile selections	8	24

The observed parent-process success rates were 75 percent for Ubuntu and 91.7 percent for Debian within their stated limits. These numbers describe the exact lab builds in the manifest. They are not a claim about arbitrary WordPress installations, package updates, pool sizes, or traffic patterns.

The reduced machine-readable record and source result hashes are in `profile_driver_qualification.json`.

Why attempts can miss

The trigger needs a useful heap neighbour in the reserved worker. A returned leak can identify the correct build and module bases while that local heap layout is still unsuitable. Structural checks reject known unsafe layouts before sending a trigger. Other compatible layouts can be attempted within the profile's fixed limit.

Ubuntu needs a separate thread-layout calibration after profile selection. Debian can reuse libc and MagickCore bases while the same PHP-FPM parent process remains alive. A parent-process restart invalidates those cached bases and requires another leak.

Test boundary

The target containers used the package versions and FPM configuration declared in the manifest. Qualification did not add a target-side helper or executable web file. The exploit interacted with WordPress through its media and HTTP routes. Public package files and debugger traces were used offline during development to calculate and verify build-relative constants.

Reproduce the local checks

```
python3 -m unittest discover -s tests -p 'test_*.py'
python3 tools/audit_supported_path_constants.py
sha256sum -c SHA256SUMS
```

Build and smoke-test both target images with:

```
./scripts/build_ubuntu.sh
./scripts/build_debian.sh
./scripts/smoke_container.sh ubuntu 8877
./scripts/smoke_container.sh debian 8878
```

`docs/returned-pixel-fingerprint-study.md`

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Returned-pixel native build fingerprints

This study asked whether bytes recovered from WordPress-generated image derivatives can identify the exact native build required by the exploit. A complete pointer pattern is exact when present. Its absence is treated as unknown because the relevant bytes are not returned on every attempt.

What the response contains

The disclosure image causes native heap bytes to appear in image pixels. A magnification stage turns each source byte into a uniform 8 by 8 luma block. The client median-decodes those blocks after WordPress saves the derivative as JPEG, producing a 32 KiB byte window.

The selector scans aligned canonical pointers in that window. Ubuntu can expose this libheif relationship:

```
response p+0x00: libheif + 0x18bf30
response p+0x10: libheif + 0x18bed0
```

Debian can expose any of these exact patterns:

```
response p+0x00: libheif + 0x18ed98
response p+0x10: libheif + 0x18da78 or 0x18ddf0

response p+0x00: libheif + 0x18de78
response p+0x10: libheif + 0x18cf58

response p+0x00: libheif + 0x18e9d8
response p+0x40: libheif + 0x155236
```

Every pointer in a pattern must produce the same page-aligned libheif base. The offsets are backed by the exact public ELF files listed in the profile manifest. The returned pointers supply runtime values, while offline ELF analysis only labels their build-relative offsets.

Decision rules

- One complete pattern for one build is positive identification.
- No complete pattern means unknown.
- Competing builds or bases mean ambiguous and stop the run.
- The HTTP `Server` header cannot select a build.
- A successful build match must still agree with the libc evidence.

The selector never substitutes an operating-system guess or a default profile for missing pointer evidence.

Validation

The Debian validation matched three of 36 returned images across two of three fresh parent processes. Every match selected the intended Debian artifact. There were no wrong or ambiguous selections. The remaining parent process was reported as unknown.

The historical Ubuntu pair was absent from the held-out images. It remains a valid positive pattern when present, but its availability depends on heap state. Ubuntu 24.04 remains an explicit negative because its stock libheif version predates the affected `unci` code and no disclosure corpus exists for that stack.

The machine-readable evidence is in `returned_pixel_fingerprint_study.json` and `remote_profile_selector_validation.json`.

Offline analyzer

```
python3 tools/analyze_returned_pixel_fingerprints.py \
  --sample response=decoded-response.bin
```

The analyzer accepts decoded returned-image bytes and the fixed candidate set. It does not query target processes or target-side metadata.

Archived from https://github.com/FORTBRIDGE-UK/libheif-unci-wordpress-rce. Rendered offline from the local Markdown copy.